

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 6 Summer 2025 (Regular/Supplementary)

Course: Bachelor of Technology (Mechanical Engineering) Branch: Engineering and Technology Semester: SEMESTER - 6

Subject Code & Name: BTMOE605C – Energy Conservation and Management

Max Marks: 60

Date: 10/07/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q. 1	Objective type questions. (Compulsory Question)				12
1	Which of the following is a key challenge associated with renewable energy sources?				1
	a. Cause higher pollution levels	b. constantly available	c. High initial investment costs	d. produces no energy	
2	Energy conservation typically involves				1
	a. Using more advanced equipment to reduce energy use	b. Maintaining the same equipment but improving its efficiency	c Reducing or cutting down on the overall energy consumption	d. Installing smart meters to monitor energy use	
3	The objective of Energy Management includes ____				1
	a. minimizing energy cost	b. minimizing waste	c. minimizing environmental degradation	d. All the above	
4	NPV considers which of the following?				1
	a. Time value of money	b. Cash flows over the life of the project	c. Risk associated with future cash flows	d. All the above	
5	In electrical insulation, which of the following is a good insulator?				1
	a. Copper	b. Iron	c. Aluminium	d. Rubber	
6	A flue gas analyser is used in energy audits to measure				1
	a. Indoor air quality	b. Fuel consumption	c. Emissions from combustion	d. Humidity in air	
7	What is the main benefit of Waste Heat Recovery (WHR) in industrial systems?				1
	a. Increased cooling load	b. Increased noise levels	c. Improved energy efficiency and reduced fuel consumption	d. Increased electricity generation only	
8	The economic thickness of insulation is determined based on				1
	a. Material colour	b. Cost of insulation vs. energy savings	c. Ease of installation only	d. Heat transfer coefficient alone	
9	In which of the following co-generative cycle electric power is produced first and then the				1

	remaining heat is used for the manufacturing sector?				
	a. Energy Conservation cycle	b. Bottoming cycle	c. Topping cycle	d. Regeneration cycle	
10	The payback period is defined as				1
	a. Time taken to double the investment	b. Time taken to recover the initial investment from savings	c. Total profits after investment	d. Annual profit divided by cost	
11	In furnaces, heat loss can be reduced by				1
	a. Increasing draft	b. Using high excess air	c. Maintaining proper insulation and sealing	d. Using oversized burners	
12	A load curve shows the variation of				1
	a. Frequency with time	b. Voltage with distance	c. Load demand with time	d. Power factor with temperature	
Q. 2	Solve the following.				12
A)	Explain in brief energy conservation techniques in Boilers & Furnaces.				6
B)	Define Energy Management. State the basic principles & benefits of energy management.				6
Q.3	Solve the following.				12
A)	Define a. Commercial & Non-commercial Energy b. Renewable & Non-renewable Energy				6
B)	What is cogeneration? Mention its applications.				6
Q. 4	Solve Any Two of the following.				12
A)	Explain use of various instruments used for Energy Auditing.				6
B)	Define a. Payback Period b. Net Present Value c. Internal Rate of Return				6
C)	A machine will cost Rs. 60,000 to purchase. Annual operating & maintenance costs will be Rs. 3000. The machine will save Rs. 20,000 per year in labour costs. The salvage value of the machine after 6 years will be Rs. 10,000. Draw the cash flow diagram.				6
Q.5	Solve Any Two of the following.				12
A)	Enlist the Types of Insulations.				6
B)	What is the load factor? Explain load factor with load curve analysis				6
C)	Describe the energy conservation opportunities in small and medium scale industries				6
Q. 6	Solve Any Two of the following.				12
A)	Explain Topping & Bottoming Cycles of Cogeneration.				6
B)	Explain in brief the difference between preliminary & detailed energy audit.				6
C)	List different power factor improvement methods and explain anyone method in brief				6
*** End ***					